

Homework 2

Ravi Kini

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Exercise 1

Part (a)

p	q	$q \vee \neg q$	$p \Rightarrow (q \vee \neg q)$	$p \vee (p \Rightarrow (q \vee \neg q))$
0	0	1	1	1
0	i	i	1	1
0	1	1	1	1
i	0	1	1	1
i	i	i	1	1
i	1	1	1	1
1	0	1	1	1
1	i	i	i	1
1	1	1	1	1

The formula is a tautology in Godel logic, with the following derivation.

1		p	
2		$p \vee (p \Rightarrow (q \vee \neg q))$	$\vee I1$
3		$\neg(q \vee \neg q)$	
4		q	
5		$q \vee \neg q$	$\vee I4$
6		$\neg(q \vee \neg q)$	$R3$
7		$\neg q$	$\neg I4-6$
8		$q \vee \neg q$	$\vee I7$
9		$p \vee (p \Rightarrow (q \vee \neg q))$	$\neg E3, 8$
10		$p \Rightarrow (q \vee \neg q)$	
11		$p \vee (p \Rightarrow (q \vee \neg q))$	$\vee I10$
12		$p \vee (p \Rightarrow (q \vee \neg q))$	$G31-2, 3-9, 10-11$

Since the derivation uses the G3 rule, the formula is not valid in intuitionistic logic.

We consider the Kripke model $\langle \{i, j, k\}, i, \leq, V \rangle$ where we have $i \leq j \leq k$, $V(j) = \{p\}$, and $V(k) = \{p, q\}$. Since p is not verified at i , for $p \vee (p \Rightarrow (q \vee \neg q))$ to be verified at i , $p \Rightarrow (q \vee \neg q)$ must be verified at i . For that to be true, for all v such that $i \leq v$, if p is verified at v , then $q \vee \neg q$ must be verified at v . At j , p is verified. We see that q is not verified at j , so for $q \vee \neg q$ to be verified at j , $\neg q$ must be verified at j . For that to be true, for all w such that $j \leq w$, q must not be verified at w . However, q is verified at k . It then follows that $p \vee (p \Rightarrow (q \vee \neg q))$ is not verified at i . This tells us that the formula is not valid in intuitionistic logic.

Part (b)

p	q	$p \wedge \neg p$	$q \wedge \neg q$	$(p \wedge \neg p) \Rightarrow (q \wedge \neg q)$	$\neg \neg((p \wedge \neg p) \Rightarrow (q \wedge \neg q))$
0	0	0	0	1	1
0	i	0	0	1	1
0	1	0	0	1	1
i	0	0	0	1	1
i	i	0	0	1	1
i	1	0	0	1	1
1	0	0	0	1	1
1	i	0	0	1	1
1	1	0	0	1	1

The formula is a tautology in Godel logic, with the following derivation.

1	$p \vee \neg p$	
2	p	$\wedge E1$
3	$\neg p$	$\wedge E2$
4	$q \wedge \neg q$	$\neg E2, 3$
5	$(p \wedge \neg p) \Rightarrow (q \wedge \neg q)$	$\Rightarrow I1-4$
6	$\neg((p \wedge \neg p) \Rightarrow (q \wedge \neg q))$	
7	$(p \wedge \neg p) \Rightarrow (q \wedge \neg q)$	R5
8	$\neg((p \wedge \neg p) \Rightarrow (q \wedge \neg q))$	R6
9	$\neg \neg((p \wedge \neg p) \Rightarrow (q \wedge \neg q))$	$\neg I6-8$

Since the derivation does not use the G3 rule, the formula is valid in intuitionistic logic.

We consider the Kripke model $\langle \{i, j, k\}, i, \leq, V \rangle$ where we have $i \leq j \leq k$, $V(j) = \{p\}$, and $V(k) = \{p, q\}$. For $\neg \neg((p \wedge \neg p) \Rightarrow (q \wedge \neg q))$ to be verified

at i , it must be true that for all v such that $i \leq v$, if $p \wedge \neg p$ is verified at v , then $q \wedge \neg q$ must be verified at v . At j , p is verified, but since p is verified at j , $\neg p$ is not verified at j , and so $p \wedge \neg p$ is not verified at j . A similar argument holds at k . It then follows that $(p \wedge \neg p) \Rightarrow (q \wedge \neg q)$ is verified at i , and so that $\neg\neg((p \wedge \neg p) \Rightarrow (q \wedge \neg q))$ is verified at i . This tells us that the formula is valid in intuitionistic logic.

Part (c)

p	q	$p \Rightarrow q$	$p \vee (p \Rightarrow q)$
0	0	1	1
0	i	1	1
0	1	1	1
i	0	0	i
i	i	1	1
i	1	1	1
1	0	0	1
1	i	i	1
1	1	1	1

The formula is not a tautology in Godel logic. The formula is then invalid in intuitionistic logic.

We consider the Kripke model $\langle \{i, j, k\}, i, \leq, V \rangle$ where we have $i \leq j \leq k$, $V(j) = \{p\}$, and $V(k) = \{p, q\}$. Since p is not verified at i , for $p \vee (p \Rightarrow (q \vee \neg q))$ to be verified at i , $p \Rightarrow q$ must be verified at i . For that to be true, for all v such that $i \leq v$, if p is verified at v then q must be verified at v . We see that p is verified at j , but q is not verified at j . It follows that $p \vee (p \Rightarrow (q \vee \neg q))$ is not verified at i . This tells us that the formula is not valid in intuitionistic logic.

Exercise 2

Part (a)

1	$\Box(p \wedge q)$	
2	$((p \wedge q) \Rightarrow (p \wedge q)) \Rightarrow (p \wedge q)$	$\Box E1$
3	$p \Rightarrow p$	
4	$p \wedge q$	
5	$p \wedge q$	R4
6	$(p \wedge q) \Rightarrow (p \wedge q)$	$\Rightarrow I4-5$
7	$p \wedge q$	$\Rightarrow E1, 6$
8	p	$\wedge E7$
9	$(p \Rightarrow p) \Rightarrow p$	$\Rightarrow I3-8$
10	$\Box p$	$\Box I9$
11	$q \Rightarrow q$	
12	$p \wedge q$	
13	$p \wedge q$	R12
14	$(p \wedge q) \Rightarrow (p \wedge q)$	$\Rightarrow I12-13$
15	$p \wedge q$	$\Rightarrow E1, 14$
16	q	$\wedge E15$
17	$(q \Rightarrow q) \Rightarrow q$	$\Rightarrow I11-16$
18	$\Box q$	$\Box I17$
19	$\Box p \wedge \Box q$	$\wedge I10, 19$

We then see that $\Box(p \wedge q) \vdash \Box p \wedge \Box q$ is a valid argument in relevant logic.

This derivation does not satisfy the entailment restriction since $p \wedge q$ was reiterated into a subproof on lines 5 and 13.

Part (b)

1	$(p \Rightarrow q) \circ (r \Rightarrow p)$		
2	$p \Rightarrow q$		
3	$r \Rightarrow p$		
4	r		
5	p	$\Rightarrow$ E3, 4	
6	q	$\Rightarrow$ E2, 5	
7	$r \Rightarrow q$	$\Rightarrow$ I4-6	
8	$(r \Rightarrow p) \Rightarrow (r \Rightarrow q)$	$\Rightarrow$ I3-7	
9	$(p \Rightarrow q) \Rightarrow ((r \Rightarrow p) \Rightarrow (r \Rightarrow q))$	$\Rightarrow$ I2-8	
10	$r \Rightarrow q$	$\circ$ E1, 9	

We then see that $(p \Rightarrow q) \circ (r \Rightarrow p) \vdash r \Rightarrow q$ is a valid argument in relevant logic.

This derivation satisfies the entailment restriction since only formulas of the form $A \Rightarrow B$ have been reiterated into subproofs.

Exercise 3

Consider the semi-lattice $S = \{0, a, b, c\}$ and the following semi-lattice operation:

$\sqcup$	0	a	b	c
0	0	a	b	c
a	a	a	c	c
b	b	c	b	c
c	c	c	c	c

We want the formulas $p \Rightarrow (p \Rightarrow p)$ and $(q \Rightarrow q) \Rightarrow (p \Rightarrow p)$ to simultaneously fail to hold. Let $V(p) = \{a, b\}$ and $V(q) = \{a\}$.

For $p \Rightarrow (p \Rightarrow p)$ to fail at 0, there must exist some v such that $v \vdash p$ and $0 \sqcup v \not\vdash (p \Rightarrow p)$. For the latter to hold, there must exist some w such that $w \vdash p$ and $v \sqcup w \not\vdash p$. Let $v = a$ and $w = b$. We see that $a \vdash p$ and $b \vdash p$; however, $a \sqcup b = c$ and $c \not\vdash p$, so $p \Rightarrow (p \Rightarrow p)$ fails at 0.

For $(q \Rightarrow q) \Rightarrow (p \Rightarrow p)$ to fail at 0, there must exist some v such that $v \vdash (q \Rightarrow q)$ and $0 \sqcup v \not\vdash (p \Rightarrow p)$. For the former to hold, for all u such that $u \vdash q$, $v \sqcup u \vdash q$. For the latter to hold, there must exist some w such that $w \vdash p$ and $v \sqcup w \not\vdash p$. Let $v = a$ and $w = b$. We see that only $a \vdash q$, and since $a \sqcup a = a$, $a \sqcup a \vdash q$ and so the former holds. We see that $a \vdash p$ and $b \vdash p$; however, $a \sqcup b = c$ and $c \not\vdash p$, so $p \Rightarrow (p \Rightarrow p)$ fails at 0.

Exercise 4

Part (a)

p	q	$p \Rightarrow q$	$p \Rightarrow \neg p$	$(p \Rightarrow \neg p) \Rightarrow q$	q
0	0	i	i	0	0
0	i	i	i	i	i
0	1	i	i	1	1
i	0	0	i	0	0
i	i	i	i	i	i
i	1	1	i	1	1
1	0	0	0	i	0
1	i	i	0	i	i
1	1	1	0	i	1

The argument is valid in connexive logic. The following derivation works for connexive logic.

1	$p \Rightarrow q$	
2	$(p \Rightarrow \neg p) \Rightarrow q$	
3	p	
4	q	$\Rightarrow$ E1, 3
5	$\neg p$	
6	p	
7	$\neg p$	R5
8	$p \Rightarrow \neg p$	$\Rightarrow$ I6-7
9	q	$\Rightarrow$ E2, 8
10	q	TND3-4, 5-9

Part (b)

p	q	$\neg q$	$\neg(p \Rightarrow q)$
0	0	1	i
0	i	i	i
0	1	0	i
i	0	1	1
i	i	i	i
i	1	0	0
1	0	1	1
1	i	i	i
1	1	0	0

The argument is valid in connexive logic. The following derivation works for connexive logic.

1		$\neg q$	
2			p
3			$\neg q$ R1
4		$\neg(p \Rightarrow q)$	$\neg \Rightarrow$ I2-3

Part (c)

p	$p \Rightarrow \neg p$	$\neg(p \Rightarrow \neg p) \Rightarrow p$	p
0	i	0	0
i	i	i	i
1	0	1	1

The argument is valid in connexive logic. The following derivation works for connexive logic.

1	$\neg(p \Rightarrow \neg p) \Rightarrow p$			
2		p		
3		p	R2	
4		$\neg p$		
5			p	
6			$\neg \neg p$	$\neg \neg$ I5
7		$\neg(p \Rightarrow \neg p)$		$\neg \Rightarrow$ I5-6
8		p		$\Rightarrow$ E1, 7
9	p			TND2-3, 4-8